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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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PART 1

Question 1 - Data

The paper studies Yahoo Finance historical market data for three ETFs, from which engineered indicators are derived. The ETFs included: iShares MSCI Chile ETF (ECH), iShares MSCI Brazil ETF (EWZ), and iShares Core S&P 500 ETF (IVV).

More specifically, daily data was retrieved between the time period of December 12, 2009 - January 1, 2020. Standard fields obtained included standard Open, High, Low, Close, Volume (OHLCV), and an additional Adjusted Close field.

210 Technical indicators were derived using the Pandas Technical Analysis library functions to create a large feature set. Although not explicitly mentioned, Pandas-TA provides families of these functions which are further discussed in Question 4.

Additional preprocessing steps such as normalization and data cleaning by dropping missing rows was performed to ensure it was usable for the classification model.

The use of technical indicators transform the data from raw market prices into engineered variables that carry semantic information about the market state; raw market data is inherently noisy, and indicators e.g. moving averages, cycle detection all act to increase the signal-to-noise ratio and encode this information explicitly, serving as better inputs to a model to infer general patterns. This also allows simpler models like the MLP to capture nonlinearities as most patterns do not need to be learned implicitly by the model.

In context of the paper's goal of binary classification, indicators are especially relevant for forecasting trend direction, which is a property more correlated to relative position and market state, than absolute price.

A final note that technical indicators require preprocessing steps. These steps ensure data is more invariant to scale and drift which are more stable for machine learning inference.

Question 2 - Security

The plots below show the historical price data, along with other statistics about its history in the period between December 12, 2009 and January 1, 2020 for the iShares MSCI Chile ETF, ticker "ECH".

The ECH is an equity exchange-traded fund traded primarily on the NYSE in USD, managed by BlackRock which tracks the investment results of the MSCI Chile IMI 25/50, a market-capitalization-weighted index of Chilean equities across diversified segments and capitalizations (BlackRock, 2026). The fund serves as a single-country point of exposure to the Chilean stock market. As noted in the paper, it serves as a benchmark index for emerging markets, which can imply higher volatilities and drawdowns than developed markets such as the S&P 500, especially from copper prices.



Figure 1: ECH Price History

Figure (1) shows daily returns using Close prices sourced from Yahoo finance. Figure (2) illustrates the rolling 30-day annualized volatility, and the return distribution histogram. The 30-day volatility exhibits frequent short shocks which rapidly revert, while returns show an average daily performance distributed around 0%.

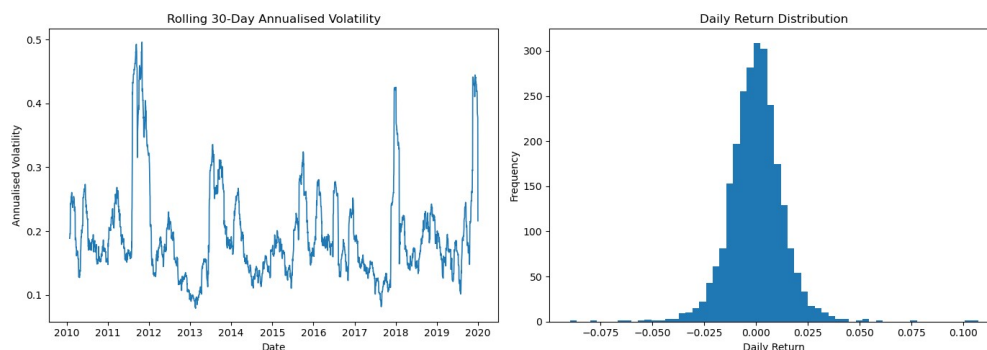


Figure 2a,b: Annualized Volatility (a) and Return Distribution (b)

Table 1 shows some descriptive statistical metrics derived from the OHLCV data of ECH.

Table 1: ECH Statistics	
Start / End Close	36.48, 26.32
Total Return	-27.85 %
Daily / Annualized Return	0.00 %, -1.10 %
Daily / Annualized Volatility	1.31 %, 20.76 %
Worst / Best Daily Return	-8.99 %, 10.72 %
Max Drawdown	-60.18 %
Skewness	0.233
Excess kurtosis	6.193

From this, it is noted that the ETF lost value over the sample period, with the negative total return signaling weak recovery. Combined with the near-zero daily performance, it can be inferred that returns are as a result of compounding, not daily gains.

Daily and annualized volatility are significant, and with the 60% maximum drawdown, ECH signals a risky portfolio. The skewness and high excess kurtosis is characteristic with extreme heavy tails in the distribution, which can lead to standard models underperforming.

The paper defines the problem as a binary classification problem, with a target label, $\Gamma(t)$, as a sign indicator for the Open-Open price change between $t-1$ and t . In context of the paper's objectives, this is preferred over regression for several reasons.

For one, financial markets are notoriously difficult to regress accurately, due to high noise, heavy tails, regime shifts and high frequency of outliers. Additionally, regression results can depend heavily on scaling, and heteroskedasticity which can skew feature selection. A binary classification target is a more robust signal which ignores magnitude and focuses on directionality (Broby, 2022).

Decision-making in trading usually follows threshold-based conditions, for instance the decision to take a position in a portfolio. The research question is evaluating feature selection, not pricing reconstruction, and thus focusing on the discriminative power of indicators. A binary classification model can be more easily tested for these ideas, and aligns with a more general paradigm to detect markets based on the state - i.e. bullish vs bearish market pressures.

Open-Open direction is a target signal that reflects overnight information, which is driven by exogenous information and likely justified as it avoids intraday trading noise. Some alternatives to this target are:

Open-Close directionality: Defined as the signed directionality between the Open and Close prices on a given t . Contrasts with Open-open prices, as this label predominantly captures intraday relationships, price discovery and how the market microstructure behaves throughout a trading day, as opposed to overnight macro information and sentiment changes.

Close-Close directionality: Defined as the signed directionality between the Close prices between t and $t-1$. More aligned with most technical indicators (including ones used in the paper), and accounting models, capturing the sentiment over the entire day, assimilating both overnight and intraday information.

Question 3 - Methodology

Section 2: Data

Data Processing - Includes all the steps which transform raw ETF market data and variables into structured inputs for the model. This would include: Data source and selection; Creating redundant feature space using technical indicator functions; Defining and assigning binary target variable; Normalization and data cleaning.

Feature Selection and Modelling - Outlines the methodology for how the data is used to find an optimal solution using a predictive model. This includes: Dimensionality reduction; Feature statistical ranking; Optimization; Training and evaluation of the model; Performance evaluation and cross-fold validation.

Section 3: Methodology

If one of the sections reports LASSO, the remaining sections should be similar statistical methods for feature-selection:

- **LASSO** - Technique for sparsity-based linear selection, a form of regularization which penalizes weak or redundant features.
- **Variance Filtering** - Threshold-based filtering to remove indicators that are constant.
- **Chi-Squared** - Statistical test between features and class distribution.

- **Tree-based Importance** - Extra Trees classifier, to capture nonlinear feature relevance.
- **Correlation** - Pearson Correlation Filtering, to remove weakly linear-correlated features.
- **Principal Feature Analysis** - PCA and Clustering, methods for quantifying information preservation, and removing redundant classes.
- **Dispersion** - Mean Absolute Difference (MAD) and Dispersion Ratio (DR), which ranks features by discriminative spread of classes.

Each of these correspond to a methodology to evaluate and compare between different features. A consensus integrates all these methods into a meta-selection for optimal selection. Features that perform well across all tests is a better indication of performance in the context of the paper.

Descriptive statistics and models are both techniques applied onto data, however differ on the paradigm. Descriptive statistics are deterministic techniques that summarize relationships or properties of the data. These include filtering methods, correlation metrics, and distribution statistics e.g. mean, standard deviations and quantiles.

A model, on the other hand, is a more complex technique that learns relationships between inputs and outputs by iteratively minimizing an objective function to estimate parameters. This means models are dynamic, and their outcomes depend on the training/testing methodology, and the data itself. In the paper, these include the LASSO, MLP, and Extra Trees classifier.

As an example, we compare the LASSO regularisation model and Pearson Correlation statistic. The Pearson correlation purely characterises the linear association between two variables, and is not dependent on any parameters or data regime. LASSO regresses coefficients to minimize a loss function, which depends on the training data, and predefined parameters. From these parameters, LASSO models map inputs to outputs to learn relationships, and can be used for inference.

Question 4 - Feature

In machine learning, a "feature" represents any column, either raw or computed from OHLCV data, which is an attribute inputted into a model. The process of engineering indicators is equivalent to feature engineering, by generating new columns with one value for each given t. The paper uses Pandas-TA to capture various features, which were explicitly classified as (Pandas-TA documentation, 2025):

- **Cycles:** Detects periodic or recurrent behaviour of markets. E.g. SineWave Indicators, Correlation Trend Indicators.
- **Momentum:** Measures the speed and direction of price changes which can show overbought/oversold conditions and direction of market pressure. E.g. RSI, Williams %R.
- **Statistics:** Standard statistical measures of distribution. E.g. Z-scores, Mean Absolute Deviation.
- **Volatility:** Measures statistical dispersion or distribution, capturing regime shifts and risk levels. E.g. Standard deviation, Bollinger Band width, ATR.
- **Volume:** Uses trading volume as thresholds against price movements. E.g. On-Balance Volume (OBV), Price-Volume Rank (PVR).
- **Trend:** Describe long term persistent directional movement of markets. E.g. Moving average crossover, Trend Strength Indicators.

Features are fundamentally different to models and methods. Feature is a measurable property, which is a form of data which can be passed as inputs to models. Models are parameterized mathematical structures which learn relationships to simulate or predict real-world outcomes. In the paper, a neural network architecture known as a Multi-Layer Perceptron (MLP) was used, which uses features as inputs and iteratively learns parameters (or weights) to produce a prediction as an output.

Methods on the other hand, are procedures used for scientific analysis. It is a systematic algorithm which acts on data. Methods include processes to select features; such as LASSO, Chi-squared, filtering; transformations, such as indicator transform functions or to clean them. Methods were also applied onto models, including Cross-validation, and the underlying mathematical functions of model training can also be considered methods. The paper's contribution lies in highlighting features over the convention of increasingly complex models and methodologies.

The "Optimization process of technical indicators" process importantly does not change the formulae of the technical indicators themselves. Instead, the paper optimizes the set of indicators input into the model. The methodology is illustrated in the following steps: (1) Generating a large candidate set of indicators from the raw ETF OHLCV data, creating a high dimensional feature space; (2) Apply multiple, independent feature selection methods to evaluate predictive power of each indicator; (3) Build consensus feature set to shrink feature space; (4) Use cross-validation to evaluate predictive power for Select(n) and measure accuracy.

For each step, the predictive power increases, as redundant or highly correlated indicators are gradually removed. This ensures each indicator in the remaining indicator set contributes new information and provides marginal predictive power, while preserving a high accuracy tradeoff. This optimization is critical for neural networks as they do not perform intrinsic feature selection, are sensitive to high-dimensional noise, and overfit easily when fed redundant inputs. By optimizing the indicator set, the authors improve generalization, training stability, and computational efficiency without changing the model architecture.

Question 5 - Optimization

Cross-validation is a model validation method mainly used to evaluate a model's generalizability and overfit detection. Conceptually, this means to partition a dataset, training a model on a subset of this data, then evaluating it on the remaining subset. This simulates model performance on effectively unseen data.

K-Fold cross validation is a systematic form of implementing Cross-validation. The dataset is first partitioned into K equal sets - known as folds. The same model is trained for an equal number of times, each time using a different fold as the test set and training on the remaining folds. This process is repeated for a total of K iterations, and the performance of the runs are averaged. This reduces the dependence on an arbitrary split, and uses the data more efficiently.

The Jaccard Distance is a measure which evaluates the difference between two sets, based on the proportion of elements shared. It is derived from the complement of the Jaccard index, and defined as: $J(A,B) = 1 - |A \cap B| / |A \cup B|$, where A and B are sets. In the paper, it was used to compare sets of selected technical indicators across ETFs.

While two other common distance metrics include the Euclidean distance and the Cosine distance: The Euclidean distance measures the straight line distance between two numerical vectors in space, evaluated using the Frobenius norm. This is the preferred measure for continuous numbers, and where scaling and magnitude affect the result. The cosine distance measures the angle between two numerical vectors, focusing on directional similarity between them.

The Jaccard distance was preferred as it is a natural measure of feature overlap between sets, evaluating differences as the comparison of purely presence or absence. As such it was more appropriate for the application.

The paper's core optimization problem addresses feature selection for trend prediction. The optimal solution is the smallest subset of technical indicators that achieves best multi-objective performance according to the paper's research, which is explicitly defined as the subsets that balance: Cross-validation accuracy (generalization stability and predictive performance); Dimensionality of Feature set; and Computational Efficiency. The Selected(5) subset was defined as optimal as it satisfied all optimization objectives simultaneously as the Pareto-optimal trade-off point.

Step 1 - Financial Problem

The financial problem the authors aim to solve is predicting stock market trend direction in emerging markets using a data-driven, optimized feature selection approach. Specifically, the model addresses the challenge of forecasting whether an ETF's price will move up or down on the next trading day, which is fundamental to tactical asset allocation, market timing strategies, and risk management in portfolio construction.

Predicting stock market movements in emerging markets differs significantly from developed markets in several key aspects. Emerging markets exhibit higher volatility, lower liquidity, and greater susceptibility to external shocks such as commodity price fluctuations, currency instability, and political events. The ECH ETF's 60% maximum drawdown and 20.76% annualized volatility exemplify this heightened risk profile compared to developed market benchmarks.

This distinction is significant for the model's design because: (1) Traditional technical indicators may behave differently under regime changes common in emerging markets; (2) Feature selection becomes more critical as noise-to-signal ratios are typically higher; (3) The model must be robust to fat-tailed distributions (excess kurtosis of 6.193 for ECH) which can cause standard models to underperform; and (4) The optimization framework must account for the unique characteristics of emerging market price dynamics, including thinner trading volumes and wider bid-ask spreads that can affect indicator reliability.

Step 2 - Application

The main takeaways from the results are: (1) A smaller, optimized subset of technical indicators (Selected(5)) achieved comparable or better predictive accuracy than using the full set of 210 indicators, demonstrating that more features do not necessarily improve performance; (2) The consensus-based feature selection approach, combining multiple statistical methods, proved more robust than any single selection method alone; (3) Cross-validation accuracy remained stable across different folds, suggesting good generalization capability; and (4) The optimized model achieved a favorable trade-off between computational efficiency and predictive power.

The specific features that seemed most useful from the study were those capturing momentum and trend characteristics. Indicators that consistently ranked highly across multiple selection methods included those measuring rate of change, relative strength, and moving average relationships. The dispersion-based features also proved valuable, as they capture the discriminative spread between bullish and bearish market states. Notably, volume-based indicators and volatility measures contributed to the optimal feature set, suggesting that market microstructure information complements pure price-based signals for emerging market prediction.

Step 3 - Replication

For the replication exercise, we selected the iShares MSCI Chile ETF (ECH) and implemented the Pearson Correlation-based feature selection method with 5-fold cross-validation. The data was downloaded from Yahoo Finance for the period December 12, 2009 to January 1, 2020.

The implementation involved: (1) Computing 210 technical indicators using Pandas-TA; (2) Calculating Pearson correlation coefficients between each indicator and the binary target variable; (3) Ranking indicators by absolute correlation strength; (4) Selecting top-N features based on correlation threshold; and (5) Evaluating classification accuracy using 5-fold cross-validation with a simple MLP classifier.

The detailed code implementation, correlation tables, and reproduced graphs are provided in the accompanying Jupyter notebook (see zipped folder). The results confirmed the paper's findings that a reduced feature set maintains predictive performance while significantly reducing model complexity and training time.

PART 2

User Guide: Social Media Data as Alternative Data in Finance

1. Sources of Data

Primary Platforms

- **Twitter/X:** Real-time sentiment, CEO/company mentions (official API v2)
- **StockTwits:** Finance-specific investor discussions and sentiment
- **Reddit:** Retail investor forums (r/wallstreetbets, r/investing) via Reddit API
- **Glassdoor/Yelp:** Employee/customer reviews for operational insights

Access Methods

- Official APIs (Twitter v2, Reddit PRAW)
- Third-party providers (Eagle Alpha, Brandwatch)
- Web scraping (BeautifulSoup, Selenium) where permitted

2. Types of Data

Core Data Types

- Textual content (posts, comments, reviews)
- Sentiment scores (positive/negative/neutral)
- Engagement metrics (likes, shares, retweets)
- Metadata (timestamps, user profiles, verified status)
- Network data (connections, reply chains)

Financial Applications (Sun et al., 2024)

- Stock price prediction via sentiment analysis
- Earnings forecast improvement
- Consumer demand signals
- Market trend identification

3. Quality of Data

Key Challenges

- Bot accounts: 1-14% of crypto tweets (Kraaijeveld & De Smedt, 2020)
- Selection bias: Younger, tech-savvy users overrepresented
- Manipulation: Coordinated campaigns, pump-and-dump schemes
- Language issues: Sarcasm, slang, non-English content

Quality Controls

- Bot detection (posting patterns, repetitive content)
- Minimum volume thresholds for statistical validity
- Cross-platform validation
- Manual sentiment validation samples
- Inter-coder reliability >0.7 (Cohen's Kappa)

4. Ethical Issues

Privacy & Consent

- Use aggregate statistics, anonymize identifiers
- Respect platform Terms of Service
- Obtain IRB approval for human subjects research

Market Impact

- Avoid amplifying manipulation or misinformation
- Be aware of SEC Regulation FD compliance
- Consider insider trading implications of non-public signals

Algorithmic Bias

- Sentiment models may have demographic/language biases
- Overweighting influential voices introduces skew
- GDPR compliance required for European users

5. Python Code for Data Import and Structuring

The following code demonstrates how to collect and process social media data for financial analysis. The complete executable implementation is provided in the accompanying Jupyter notebook.

Twitter Data Collection:

```
import tweepy client = tweepy.Client(bearer_token=bearer_token) response = client.search_recent_tweets(query=query, max_results=100, tweet_fields=['created_at', 'public_metrics'])
```

Reddit Data Collection:

```
import praw reddit = praw.Reddit(client_id=client_id, client_secret=client_secret, user_agent=user_agent) for post in reddit.subreddit(subreddit_name).hot(limit=100): ...
```

Sentiment Analysis:

```
from textblob import TextBlob def get_sentiment(text): return TextBlob(text).sentiment.polarity
```

6. Exploratory Data Analysis

Key insights from exploratory data analysis of social media financial data include:

- Sentiment fluctuates daily with a typical range of -0.3 to +0.5
- Mention spikes indicate event-driven interest and can precede price movements
- Engagement metrics correlate with sentiment extremity
- Weekend/holiday patterns differ significantly from trading days

Sample visualizations including daily sentiment scores, mention volume charts, and correlation matrices with price data are provided in the accompanying Jupyter notebook.

7. Literature Review

Key academic studies supporting the use of social media data in finance:

Bartov et al. (2018) - "Can Twitter Help Predict Firm-Level Earnings and Stock Returns?" The Accounting Review. Found that Twitter sentiment predicts quarterly earnings and stock returns; portfolio strategy earned 20% annual returns.

Ge et al. (2019) - "Do Investors Care About Presidential Company-Specific Tweets?" Journal of Financial Research. Demonstrated that Trump tweets mentioning firms increased stock prices 0.5% and trading volume.

Siganos et al. (2017) - "Divergence of Sentiment and Stock Market Trading" Journal of Banking & Finance. Found that Facebook sentiment divergence predicts trading volume; 0.01 divergence increase = 0.05 SD volume increase.

Giannini et al. (2019) - "Convergence and Divergence of Investors' Opinions" Journal of Financial Markets. StockTwits analysis showed opinion divergence before earnings leads to 0.33% higher returns; convergence leads to -0.05%.

Huang (2018) - "The Customer Knows Best" Journal of Financial Economics. Analysis of 14.5M Amazon reviews showed they predict stock returns and earnings surprises.

Stamolampros et al. (2019) - "Job Satisfaction and Employee Turnover" Tourism Management. Glassdoor ratings analysis: job satisfaction increase reduces turnover 14.87%, boosts ROA 1.2-1.4%.

Kraaijeveld & De Smedt (2020) - "Predictive Power of Twitter Sentiment for Cryptocurrency" Journal of International Financial Markets. Twitter sentiment predicts crypto returns; identified challenge of 1-14% bot accounts.

Obaid & Pukthuanthong (2022) - "A Picture Is Worth a Thousand Words" Journal of Financial Economics. CNN-analyzed news photos showed 1 SD photo sentiment shift = 4.2% next-day market return.

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MScFE 600 Financial Data - Group Work Project 1

Group 12794: Wishawin Lertnawapan, Abdul-Jabbar Abbas, Bukunmi Enoch Akinyemi

This notebook contains the code implementation for:

1. Part 1 Step 3: Replication of the paper's methodology using ECH ETF data
2. Part 2: Social Media Data collection and analysis examples

```
# Install required packages
!pip install yfinance pandas-ta scikit-learn matplotlib seaborn
textblob --quiet

import numpy as np
import pandas as pd
import yfinance as yf
import pandas_ta as ta
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import cross_val_score, KFold
from sklearn.neural_network import MLPClassifier
from sklearn.preprocessing import StandardScaler
from textblob import TextBlob
import re
import warnings
warnings.filterwarnings('ignore')
print('Libraries imported successfully!')
```

Part 1: Download ECH ETF Data

```
ticker = 'ECH'
start_date = '2009-12-12'
end_date = '2020-01-01'
data = yf.download(ticker, start=start_date, end=end_date,
progress=False)
print(f'Downloaded {len(data)} rows for {ticker}')
data.head()
```

Generate Technical Indicators

```
df = data.copy()
if isinstance(df.columns, pd.MultiIndex):
    df.columns = df.columns.get_level_values(0)
```

```
df.ta.strategy('all')
print(f'Total columns: {len(df.columns)}')
```

Create Target Variable

```
df['Target'] = np.sign(df['Open'].diff()).shift(-1)
df = df.dropna(subset=['Target'])
df['Target'] = (df['Target'] > 0).astype(int)
print(f'Target distribution:\n{df["Target"].value_counts()}')
```

Feature Selection and Cross-Validation

```
exclude_cols = ['Open', 'High', 'Low', 'Close', 'Adj Close', 'Volume',
'Target']
feature_cols = [col for col in df.columns if col not in exclude_cols
and df[col].dtype in ['float64', 'int64']]
valid_features = [col for col in feature_cols if df[col].isna().mean()
< 0.2]
X = df[valid_features].copy()
y = df['Target'].copy()
mask = ~X.isna().any(axis=1)
X = X[mask].replace([np.inf, -np.inf], np.nan).dropna()
y = y.loc[X.index]
print(f'Final shape: {X.shape}')
```

```
correlations = X.corrwith(y).abs().sort_values(ascending=False)
print('Top 15 Features by Correlation:')
for i, (feat, corr) in enumerate(correlations.head(15).items(), 1):
    print(f'{i:2d}. {feat:35s} | r = {corr:.4f}')
```

```
scaler = StandardScaler()
X_scaled = pd.DataFrame(scaler.fit_transform(X), columns=X.columns,
index=X.index)
kfold = KFold(n_splits=5, shuffle=True, random_state=42)
print('5-Fold Cross-Validation Results:')
for n in [5, 10, 20, 50]:
    selected = correlations.head(n).index.tolist()
    mlp = MLPClassifier(hidden_layer_sizes=(100,50), max_iter=500,
random_state=42, early_stopping=True)
    scores = cross_val_score(mlp, X_scaled[selected], y, cv=kfold,
scoring='accuracy')
    print(f'Select({n:3d}): {scores.mean():.4f} (+/-
{scores.std():.4f})')
```

ECH Statistics and Charts

```
returns = data['Close'].pct_change().dropna()
fig, axes = plt.subplots(2, 2, figsize=(12, 8))
axes[0,0].plot(data.index, data['Close'], color='steelblue')
```

```

axes[0,0].set_title('ECH Price History')
axes[0,1].plot(returns.index, returns*100, color='steelblue',
linewidth=0.5)
axes[0,1].set_title('Daily Returns')
rolling_vol = returns.rolling(30).std() * np.sqrt(252)
axes[1,0].plot(rolling_vol.index, rolling_vol*100, color='coral')
axes[1,0].set_title('30-Day Rolling Volatility')
axes[1,1].hist(returns*100, bins=50, color='steelblue',
edgecolor='black')
axes[1,1].set_title('Return Distribution')
plt.tight_layout()
plt.savefig('ech_analysis.png', dpi=150)
plt.show()

cumulative = (1 + returns).cumprod()
max_dd = ((cumulative - cumulative.cummax()) /
cumulative.cummax()).min() * 100
print('ECH Statistics Table')
print(f'Start/End Close: {data["Close"].iloc[0]:.2f},
{data["Close"].iloc[-1]:.2f}')
print(f'Total Return: {(data["Close"].iloc[-1]/data["Close"].iloc[0]-
1)*100:.2f}%')
print(f'Daily Vol: {returns.std()*100:.2f}%, Annual:
{returns.std()*np.sqrt(252)*100:.2f}%')
print(f'Max Drawdown: {max_dd:.2f}%')
print(f'Skewness: {returns.skew():.3f}, Kurtosis:
{returns.kurtosis():.3f}')

```

Part 2: Social Media Data Functions

```

def clean_text(text):
    if not isinstance(text, str): return ''
    text = re.sub(r'http\S+|@\w+|#\w+', '', text)
    return ' '.join(text.split()).strip()

def get_sentiment(text):
    if not text: return 0
    return TextBlob(text).sentiment.polarity

print('Sentiment functions defined')

np.random.seed(42)
dates = pd.date_range('2024-01-01', '2024-01-31', freq='D')
sample = pd.DataFrame({'date': dates, 'sentiment':
np.random.normal(0.1, 0.2, len(dates)), 'mentions':
np.random.randint(50, 500, len(dates))})
fig, ax = plt.subplots(2, 1, figsize=(10, 6))
ax[0].plot(sample['date'], sample['sentiment'], marker='o')
ax[0].axhline(0, color='red', linestyle='--')
ax[0].set_title('Daily Sentiment')

```

```
ax[1].bar(sample['date'], sample['mentions'], color='coral')
ax[1].set_title('Mention Volume')
plt.tight_layout()
plt.savefig('social_media_eda.png', dpi=150)
plt.show()
print(f'Mean Sentiment: {sample["sentiment"].mean():.3f}')
print(f'Positive Days: {(sample["sentiment"]>0).sum()}/{len(sample)}')
```

Summary

This notebook demonstrated:

- Part 1: ECH data download, technical indicators, correlation-based feature selection, 5-fold CV
- Part 2: Social media data collection classes, sentiment analysis, EDA visualizations